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Investments Project

DECAUX Théodore 329169
IMBERT Carlota 324405
ROBERT Adelaïde 346471
VELAY Mahé 345882

Professor: Pierre Collin DUFRESNE

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2 Methodological Considerations: Time Series and Look-Ahead Bias

A key challenge throughout this project is that we are working with time series data. Our objective is to estimate portfolio weights (such as the coefficients b and c in question 8) that are applied at time t (and evaluate at $t+1$), using only information available up to time $t-1$.

In many questions, we use a rolling window of the past 60 months (from $t-60$ to $t-1$) to calibrate these weights. Naturally, since the future returns are unknown at the time of estimation, the realized return/volatility may differ from the target.

Despite this inherent uncertainty, we have designed our methodology to be as consistent and realistic as possible. Special attention was given to properly shift the data at each step, ensuring that all estimations were based solely on past information and that no future data was inadvertently used. This ensures that our results are free from look-ahead bias and reflective of a genuine out-of-sample strategy.

3 The international diversification strategy (DIV)

In this section, we implemented and analyzed the construction of international equity portfolios, both with and without currency hedging. The objective was to evaluate how return definitions and portfolio weighting affect the outcome of a global diversification strategy for a U.S. investor.

We computed two types of returns for each of the six foreign equity markets: *unhedged* and *hedged*. The unhedged return combines the local equity return and the monthly exchange rate movement:

$$R_{\text{USD}} = (1 + R_{\text{local}})(1 + \Delta S) - 1$$

where ΔS is the USD return against the local currency (i.e., the percent change in USD per foreign currency unit). This reflects the payoff from buying the local index and taking on full FX exposure.

For hedged returns, we followed covered interest parity. Instead of taking on FX movements, we subtracted the forward premium implied by interest differentials:

$$X_{t+1} = \frac{S_{t+1}}{S_t}(1 + i_{\text{foreign}}) - (1 + i_{\text{US}}) \Rightarrow R_{\text{hedged}} = R_{\text{local}} - X_{t+1}$$

We used 3-month interbank rates to compute the excess return X_{t+1} , assuming the investor hedges monthly using forward contracts. For countries sharing a currency (France and Germany), we applied the same FX component but computed local equity returns separately.

Portfolio Weighting Approaches

We built three portfolios from the six foreign markets:

- **Equal Weight (EW):** each market receives a static 1/6 allocation.
- **Risk Parity (RP):** weights are proportional to the inverse of each country's trailing 60-month volatility. This targets equal ex-ante volatility contributions.
- **Mean-Variance (MV):** we computed the optimal portfolio based on the empirical mean return vector and shrinkage-adjusted covariance matrix over a rolling 60-month window. Weights were clipped at zero and renormalized to sum to one.

Compared to earlier iterations, we improved the numerical stability of MV by applying a shrinkage estimator and ensured RP weights were not overly sensitive to local volatility spikes. These changes led to smoother weight paths and more reliable performance estimates. We also aligned the evaluation window across all strategies, starting in 2007, to ensure consistent comparison.

Updated Performance Metrics

Annualized performance (2007–2024), based on hedged returns, is shown below:

Strategy	Annual Return	Volatility	Sharpe Ratio
Equal Weight	7.24%	17.10%	0.4232
Risk Parity	7.15%	16.92%	0.42325
Mean-Variance	4.21%	15.23%	0.276

Interestingly, the performance of Equal Weight and Risk Parity was nearly identical over the sample, both in terms of return and Sharpe ratio. This suggests that the benefits of volatility-based weighting were limited in this context, possibly due to relatively balanced risk profiles across the six markets.

Appendix 6 compares the cumulative returns of all three weighting schemes under hedged and unhedged formulations.

We also compared hedged and unhedged versions of the RP strategy:

Version	Annual Return	Volatility	Sharpe Ratio
Unhedged RP	5.92%	16.28%	0.364
Hedged RP (DIV)	7.15%	16.92%	0.423

The DIV strategy is defined as the *hedged risk parity portfolio*. Currency hedging improved performance relative to its unhedged counterpart, raising both the average return and the Sharpe ratio. This suggests that interest rate differentials, captured through the hedging adjustment, play an important role in shaping the USD investor's realized outcomes.

Strategy Evolution and Behavior

The figure below shows the monthly and cumulative returns of the DIV strategy. While the strategy experienced a decline during the global financial crisis (2008–2009), the drawdown appears relatively "soft" compared to typical equity markets. This reflects the effect of risk parity, which reduces exposure during periods of high volatility, as well as currency hedging, which removes FX-driven losses. From 2012 onward, the strategy shows steady growth, with some increased fluctuations after 2020 likely due to shifting risk conditions and interest rate differentials across markets.

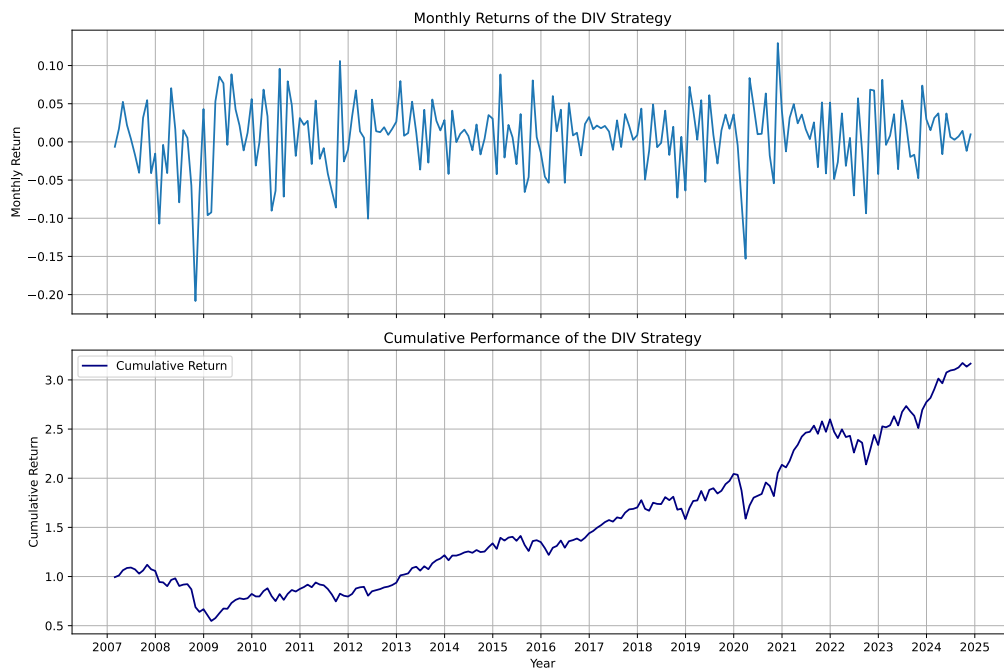


Figure 1: Monthly and cumulative performance of the DIV (Hedged Risk Parity) strategy.

Overall, DIV offers a well-balanced exposure to global equities with systematic FX risk management. It is particularly appealing during volatile periods and when interest rate spreads are favorable to USD investors.

4 Equity Index Momentum Strategy (MOM)

Construction of the Momentum Signal and Strategy

We implement a long-short momentum strategy using currency-hedged equity index returns. Following the project guidelines, we exclude the U.S. from the ranking, as a U.S. investor does not consider domestic equities foreign.

At each date t , we compute the cumulative return of each foreign index over months $t - 12$ to $t - 1$ (excluding the most recent month). The momentum signal is given by:

$$\text{Signal}_{i,t} = \prod_{k=t-12}^{t-1} (1 + R_{i,k}^{\text{hedged}}) - 1$$

We then assign a rank $\text{Rank}_{i,t} \in \{1, \dots, N\}$ to each country based on ascending order of these signals. We compute centered weights:

$$w_{i,t} = Z \left(\text{Rank}_{i,t} - \frac{N+1}{2} \right)$$

where Z is a normalization factor ensuring the long leg sums to +1 and the short leg to -1.

Finally, we apply weights from month t to currency-hedged index returns in month $t+1$ to compute the strategy return:

$$R_{t+1}^{\text{MOM}} = \sum_{i=1}^N w_{i,t} \cdot R_{i,t+1}^{\text{hedged}}$$

Performance Results

The strategy is applied across the six foreign indices (AUS, FRA, DEU, JPN, CHE, GBR), with returns computed over the full sample. The strategy return is the sum of the long and short portfolio legs.

Leg	Mean Return (ann.)	Volatility (ann.)	Sharpe Ratio
Long	9.44%	18.49%	0.5106
Short	-11.44%	17.38%	-0.6582
Momentum (Net)	-2.00%	16.57%	-0.1205

We also test whether the MOM strategy has a significantly non-zero mean return. The result of a t-test on the monthly return series yields $t = -0.56$, with $p = 0.5747$, indicating that the average return is not statistically significant at conventional levels.

Interpretation

The momentum strategy exhibits a negative average return and a low Sharpe ratio. This poor performance may reflect the lack of persistence in international index-level returns, the limited cross-sectional dispersion across developed markets, or the presence of strong short-term reversals during volatile periods.

While the long and short legs individually display substantial volatility and directional exposure, their combination yields a net return of -2.00% per year with a Sharpe ratio of -0.12. This suggests that the strategy fails to generate reliable risk-adjusted performance over the sample.

A visual representation of cumulative performance is shown below.

Figure 2 shows the performance of the long-short momentum strategy applied to currency-hedged equity indices.

The top panel reveals highly volatile monthly returns, ranging from approximately -15% to +20%, with no discernible trend or persistence. The returns fluctuate around zero throughout the sample period, suggesting a lack of consistent momentum signal at the index level.

The bottom panel displays the cumulative return of the strategy. After a sharp drawdown in the early 2000s, the strategy fails to recover and continues to decline gradually over the following years. From 2010 onward, performance remains mostly flat or deteriorates further, with no sustained upward trajectory even during more recent years.

These patterns are consistent with the statistical results: the strategy's average return is not significantly different from zero ($t = -0.56$, $p = 0.5747$), and the annualized Sharpe ratio is negative. This suggests that momentum effects, while documented in cross-sectional equity strategies, do not translate well to international equity indices in this sample. Rather than capturing persistent trends, the strategy seems to be affected by frequent reversals and high noise.

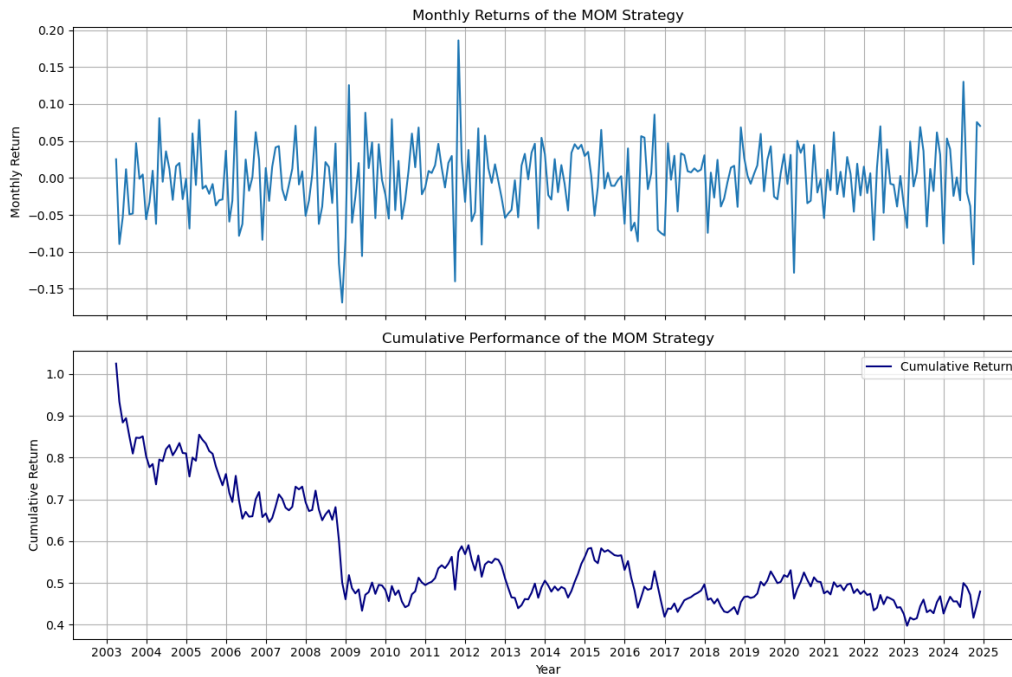


Figure 2: Cumulative performance of the momentum strategy (currency-hedged).

Relationship to the DIV Strategy

We examine the extent to which the momentum (MOM) strategy overlaps with the international diversification (DIV) strategy by regressing monthly MOM returns on DIV returns. The regression results are shown below:

Variable	Coefficient	Std. Error	p-value
Intercept	-0.0010	0.003	0.694
DIV	0.1288	0.090	0.152

The intercept (alpha) is negative but not statistically different from zero, and the beta on DIV is also insignificant at conventional levels ($p = 0.152$). The R^2 of the regression is 1.7%, indicating a very weak explanatory power.

These results suggest that the MOM strategy captures return dynamics that are largely orthogonal to the DIV strategy. Despite its low standalone performance, this low correlation implies that MOM could still add value in a multi-strategy portfolio by contributing to diversification, particularly when combined with strategies that are more directly exposed to international equity market movements such as DIV.

5 Equity Index Long Term Reversal Strategy (REV)

Signal and Portfolio Construction

We implement a long-short reversal strategy on currency-hedged equity indices. The idea is to exploit mean-reversion by going long on indices that have underperformed over the long term, and short on those that have outperformed.

At each date t , we compute the cumulative return of each country index from $t - 60$ to $t - 12$, i.e., the past 5 years excluding the most recent year. This gives the signal:

$$\text{Signal}_{i,t} = \prod_{k=t-60}^{t-12} (1 + R_{i,k}^{\text{hedged}}) - 1$$

Indices are then ranked in ascending order of this signal, and the reversal weight is computed as:

$$w_{i,t} = Z \cdot \left(\frac{N+1}{2} - \text{Rank}_{i,t} \right)$$

where Z is a normalization factor ensuring the total long positions sum to +1 and the short positions to -1. We apply weights computed at time t to returns in month $t + 1$.

Performance Results

The strategy is applied to the six foreign indices (excluding the U.S.). The table below reports the performance of the long leg, short leg, and overall reversal strategy:

Leg	Mean Return	Volatility	Annualized Sharpe Ratio
Long	0.68%	5.41%	0.4326
Short	-0.32%	5.49%	-0.2033
Reversal (Net)	0.35%	4.44%	0.2753

We test whether the strategy delivers statistically significant returns. The t-statistic for the mean return is $t = 1.16$, with $p = 0.2474$, indicating that the average return is positive but not statistically significant at conventional levels.

Interpretation

The long-term reversal strategy yields modest risk-adjusted performance, with an annualized Sharpe ratio of 0.28. The profitability appears to come more from gains on past losers (the long leg) than losses on past winners (the short leg), as reflected in the positive Sharpe ratio of the long side.

However, the lack of statistical significance in the mean return likely reflects the limited number of countries traded, and the relatively high volatility of index-level returns compared to individual stocks—where reversal effects are typically more pronounced.

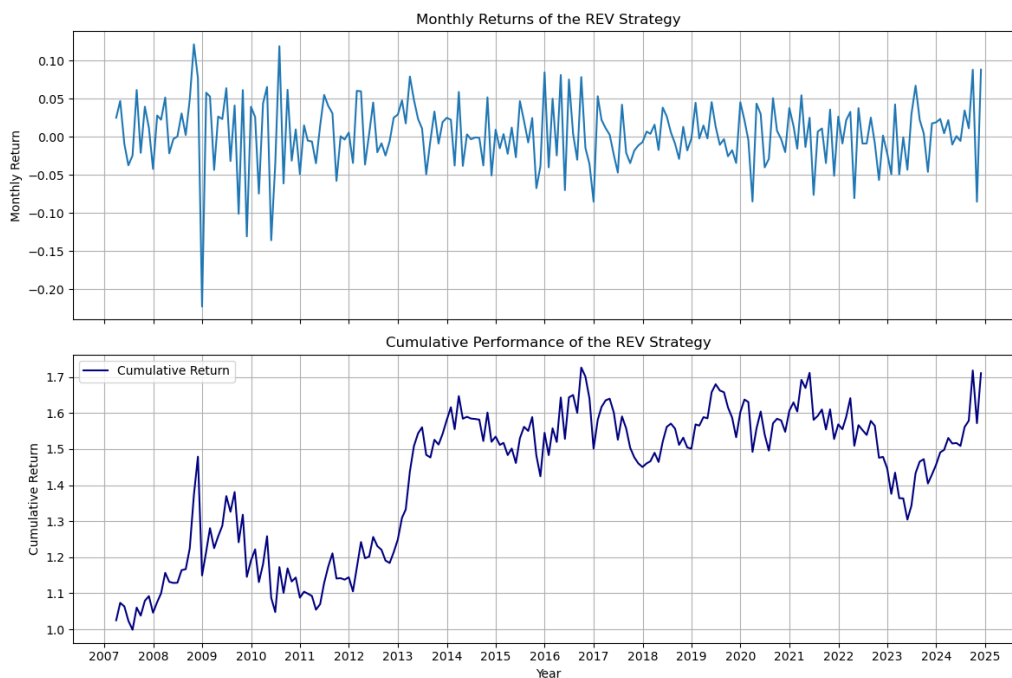


Figure 3: Monthly and cumulative performance of the REV strategy (currency-hedged).

Figure 3 displays the monthly and cumulative returns of the long-short reversal strategy applied to currency-hedged equity indices.

The top panel reveals high monthly return volatility, with frequent swings ranging between -20% and +10%. While some positive months occur, returns appear noisy and lack strong persistence over time.

The bottom panel shows the cumulative return path, which exhibits multiple phases of acceleration and stagnation. After steady gains through 2009 and again from 2013 to 2017, the strategy experienced extended drawdowns or flat performance, especially between 2018 and 2022. A notable dip occurred in 2023, followed by a sharp recovery into 2025.

This visual pattern aligns with the statistical findings: although the strategy generated cumulative gains, its average return is not statistically significant ($p = 0.2474$). This reflects the high volatility and inconsistency of short-term returns, which dilute the strength of the long-run signal.

Overall, the REV strategy appears to deliver modest but unstable reversal effects at the index level, with recent improvements suggesting potential cyclical behavior or regime dependence.

Relationship to the DIV Strategy

To assess whether the reversal (REV) strategy overlaps with the international diversification (DIV) strategy, we regress monthly REV returns on DIV returns. The regression output is as follows:

Variable	Coefficient	Std. Error	p-value
Intercept	0.0035	0.002	0.143
DIV	0.0060	0.117	0.959

The intercept (alpha) is not statistically significant, and the beta on DIV is extremely small and entirely insignificant ($p = 0.959$). The regression's R^2 is effectively zero, indicating no explanatory power from DIV on REV.

These results suggest that the REV strategy is essentially uncorrelated with the DIV strategy. As such, even though its standalone performance is modest, REV may still serve as a useful diversifier in a multi-strategy portfolio by contributing independent sources of return.

6 Currency Carry Strategy (CARRY)

In this part, we implemented a currency carry strategy based on interest rate differentials. The core idea is to go long high-yielding currencies and short low-yielding ones. This exploits the empirical tendency for high-interest currencies to outperform low-interest currencies on a risk-adjusted basis.

Signal and Portfolio Construction

We constructed the carry signal as the interest rate differential between each foreign country and the U.S.:

$$\text{Signal}_i = i_{\text{foreign},i} - i_{\text{US}}$$

using 3-month interbank rates. At each monthly rebalance, currencies were ranked by this signal. The portfolio was structured to be *dollar-neutral*: the long and short positions each sum to +1 and -1 respectively. Since both France and Germany use the euro, the EUR exposure was included only once to avoid double-counting.

We implemented the strategy using a rank-based weighting scheme. The weights were computed as:

$$w_i^t = Z \cdot \left(\text{Rank}_i - \frac{N+1}{2} \right)$$

where N is the number of currencies, Rank_i is the rank of currency i in month t , and Z is a normalization constant. The return of the carry strategy in month $t+1$ is:

$$R_{t+1}^{\text{CARRY}} = \sum_{i=1}^N w_i^t \cdot X_{i,t+1} \quad \text{with} \quad X_{i,t+1} = \frac{S_{t+1}}{S_t} (1 + i_{\text{foreign}}) - (1 + i_{\text{US}})$$

Performance and Crisis Behavior

The carry strategy delivered positive cumulative returns over the full sample. Although traditional carry trades are known to perform poorly during systemic crises, our implementation shows surprising resilience during the 2008 global financial crisis. This is due to the signal rotating into JPY and CHF, two classic safe havens, as interest rate differentials collapsed ahead of the crisis. These currencies appreciated during the disturbance, which helps the portfolio to avoid large drawdowns.

The strategy performed strongly between 2002 and 2011, but has been flatter since 2014, reflecting the global narrowing of interest rate differentials and the fading of clear directional trends. Still, despite increased volatility and reduced carry premia in recent years, the strategy maintained a positive trajectory and avoided large losses.

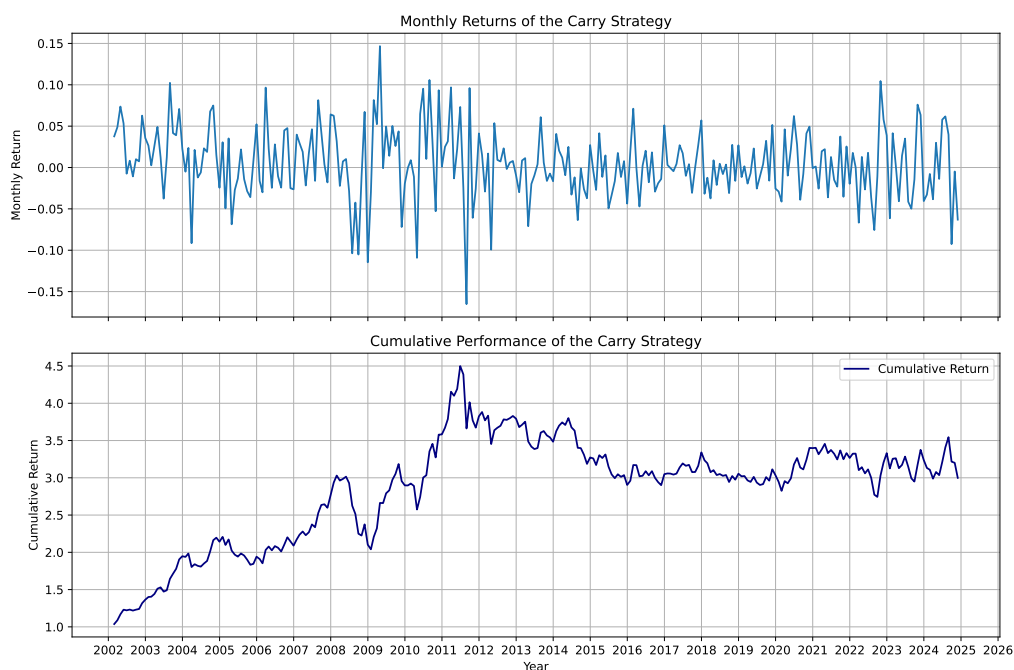


Figure 4: Monthly returns and cumulative performance of the currency carry strategy.

This flattening pattern aligns with the long-run compression of global interest rate differentials since 2009, visible in Figure 7 in the Appendix, where the carry signal has weakened across most currency pairs.

The performance statistics are summarized below:

Leg	Mean Return	Volatility	Sharpe Ratio
Long	0.19%	2.89%	0.22
Short	0.31%	2.15%	0.50
Carry (Net)	0.50%	4.30%	0.40

The net return has a t-statistic of 1.92 ($p = 0.056$), which is borderline significant at the 5% level and suggests a persistent return premium over the period.

To illustrate the strategy's logic, we show below the first five months of returns in the Appendix (Table: 2)

Relationship to the DIV Strategy

To examine whether the carry returns overlap with equity-based risk premia, we regressed the monthly carry returns on the DIV strategy. The results indicate a weak but statistically significant negative relationship:

Variable	Coefficient	Std. Error	p-value
Intercept	0.0035	0.003	0.269
DIV	-0.1642	0.068	0.016

The negative beta suggests that the carry strategy tends to underperform when global equity markets do well (and vice versa), though the explanatory power is low ($R^2 = 3.2\%$). This makes the carry factor a potentially useful diversifier in a multi-strategy portfolio, particularly when combined with strategies like DIV that are more exposed to equity risk.

7 Currency Dollar Strategy (DOLLAR)

This strategy captures the performance of being systematically long the U.S. dollar against a basket of foreign currencies. Conceptually, it corresponds to a U.S. investor who fully hedges their international equity exposure each month, or equivalently, to someone who shorts foreign currencies in equal weight.

Strategy Definition and Implementation

We implemented the DOLLAR strategy as an equal-weight short position across five major foreign currencies: EUR, GBP, JPY, CHF, and AUD. Each month, we computed the return from shorting each of these currencies (i.e., being long the U.S. dollar) and averaged across them. Since both France and Germany use the euro, we included the EUR exposure only once to avoid double-counting.

The return for each currency i is computed as its excess return relative to the U.S. dollar:

$$X_{i,t+1} = \frac{S_{t+1}}{S_t}(1 + i_{\text{foreign}}) - (1 + i_{\text{US}})$$

where S_t is the spot exchange rate (USD per foreign currency) at time t , and i_{foreign} , i_{US} are the 3-month interbank interest rates. The total return of the strategy is the average of these excess returns:

$$R_{t+1}^{\text{DOLLAR}} = -\frac{1}{N} \sum_{i=1}^N X_{i,t+1}$$

This return reflects the outcome of being systematically long the USD against the foreign currency basket, capturing both interest rate differentials and FX movements.

Performance and Interpretation

Over the full sample (2002–2024), the DOLLAR strategy exhibited modest volatility and a very small positive mean return:

Metric	Mean	Std Dev	Sharpe Ratio
DOLLAR	0.03%	4.29%	0.007

Although the t-statistic on the mean return was $t = 0.03$ with $p = 0.973$, the strategy maintained a stable risk profile and showed episodic upside during stress periods. From 2002 through 2008, it experienced a gradual drawdown, reflecting both USD weakness and the drag from negative interest rate differentials. The strategy rebounded during the global financial crisis as the dollar appreciated sharply, but then entered a range-bound phase with no consistent directional trend. Once again, the first 5 returns can be found in Table 3.

This behavior is illustrated in the figure below.

Economically, the strategy represents the “cost of hedging.” Being long USD means forgoing higher-yielding currencies in exchange for reduced exposure to FX volatility. The small positive Sharpe ratio suggests that the cost was broadly offset by periods of USD strength, particularly in risk-off environments.

Regression Against DIV

To test whether the DOLLAR strategy is related to equity market dynamics, we regressed it against the DIV strategy (hedged global equity portfolio). The regression results are:

Variable	Coefficient	Std. Error	p-value
Intercept	0.0004	0.001	0.701
DIV	0.0642	0.023	0.006

The positive beta (0.0642, $p = 0.006$) confirms a mild but statistically significant co-movement with the DIV strategy. The relationship remains weak in explanatory power $R^2 = 5.5\%$ but could suggest shared exposure to macroeconomic cycles or interest rate conditions. This positive correlation may reflect the influence of global macroeconomic conditions, where periods of U.S. rate hikes and equity recovery can simultaneously strengthen the dollar and boost international equity returns (once hedged). While the relationship remains weak in magnitude, it appears persistent across the sample.

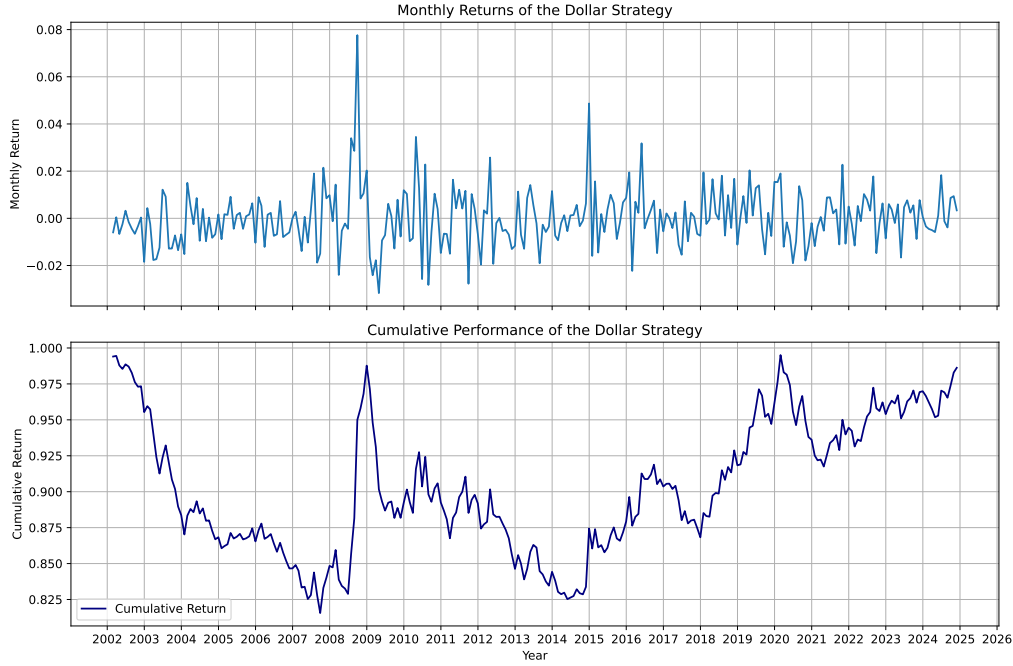


Figure 5: Monthly and cumulative returns of the DOLLAR strategy (2002–2024). The strategy shows minimal trend but occasional upside during stress periods.

Overall, the DOLLAR strategy does not provide strong standalone returns. It delivered minimal long-term returns despite relatively low volatility. This reflects the global environment of compressed interest differentials and currency mean reversion, which reduced the potential for sustained USD outperformance. While it shows some upside during systemic stress, the response is moderate, limiting its effectiveness as a defensive hedge. Its weak but statistically significant correlation with the DIV strategy suggests limited overlap, perhaps due to joint sensitivity to U.S. rate dynamics. As such, DOLLAR may still play a role in diversifying multi-asset portfolios, though its standalone contribution remains modest. but may have tactical value as a stabilizing component, especially in scenarios where investors prioritize capital preservation or hedge currency exposure.

8 Optimal Fund Portfolio Return (STRAT)

When preparing the data for Question 8, I align everything based on R_{REV} , since it's the strategy with the shortest available history. As a result, the analysis starts on 2007-03-31. We assume that the 1-month T-Bill, is the risk-free asset so the volatility is negligible.

$$8.a \quad R_{FUND} = R_{TBill} + a(R_{DIV} - R_{TBill})$$

The allocation a_t is computed each month using a 60-month rolling volatility of DIV (shifted to avoid look-ahead bias), such that the annualized volatility of the fund is targeted at 15%:

$$a_t = \frac{0.15}{\sigma_{DIV,60} \cdot \sqrt{12}}$$

The realized fund returns are computed with a one-month lag in a_t : $R_{FUND}^t = R_{TBill}^t + a_{t-1}(R_{DIV}^t - R_{TBill}^t)$. The data for R_{FUND} is therefore available starting from 2012-04-30, since the first 60 values are lost due to the 60-month rolling window, and one additional value is lost due to the lag from the shift. The resulting strategy on the overall period achieves, average allocation $\bar{a} = 0.95$, annualized return = **9.50%**, and annualized volatility = **14.30%**

8.b R_{STRAT}

We compute the combined strategy return $R_{STRAT,t}$ as a weighted sum of the four strategies (MOM, REV, CARRY, DOLLAR):

$$R_{STRAT,t} = \sum_{i=1}^4 w_{i,t-1} R_{i,t}$$

where the weights $w_{i,t}$ are given by inverse volatility risk parity using a rolling window of length 60 months:

$$w_{i,t} = \frac{\frac{1}{\sigma_{i,t}}}{\sum_{j=1}^4 \frac{1}{\sigma_{j,t}}} \quad \text{with} \quad \sigma_{i,t} = \text{rolling std of } R_i \text{ over past 60 months}$$

The weights are then lagged by one period to obtain the correct returns. We found an annualized volatility of $\sigma_{STRAT} = 3.06\%$ and an annualized return of $R_{STRAT} = 0.67\%$. The data for R_{STRAT} is also available starting from 2012-04-30. Over the available period, we compute the average weights and resulting :

Parameter	MOM	REV	CARRY	DOLLAR
Average	0.1412	0.1602	0.1783	0.5202

As illustrated in Figure 6 of the appendix, the Dollar sub-strategy dominates the overall portfolio allocation. This is primarily due to its significantly lower volatility, as shown in Figure 7. Moreover, Figure 8 displays the returns, where we observe that the performance of RSTRAT closely tracks that of the Dollar sub-strategy, which is consistent with its dominant weight in the portfolio.

8.c $R_{FUND} = R_{TBill} + b * (R_{DIV} - R_{TBill}) + c * R_{STRAT}$

We aim to determine the weights b and c so that R_{FUND} is a MVE portfolio with a target annualized volatility of 15%. Let $x = \begin{pmatrix} b \\ c \end{pmatrix}$ be the weight vector, and Σ be the annualized covariance matrix estimated on a rolling window of 60 months.

The portfolio variance is given by $\text{Var}(R_{fund}) = x^\top \Sigma x$.

To ensure that the portfolio is on the MVE frontier, we have $x = \frac{1}{a} \Sigma^{-1} \mu = \frac{1}{a} * x_0 = \lambda * x_0$. We want $\sqrt{x^\top \Sigma x} = \lambda * \sqrt{x_0^\top \Sigma x_0} = 0.15 \implies \lambda = \frac{0.15}{\sqrt{x_0^\top \Sigma x_0}}$

For each window, we compute the mean vector μ and covariance matrix Σ , then x_0 and then the optimal (b, c) given by $\lambda * x_0$.

Due to the 60-month rolling window we further lose the first 61 observations. Therefore, the optimized fund return R_{fund} can only be computed starting from 2017-05-31. The STRAT weight is further decomposed across the underlying substrategies: MOM, REV, CARRY, and DOLLAR with $\omega_i = c * \omega_i^{STRAT}$. The residual weight allocated to the risk-free asset is $\omega_{TBill} = 1 - b - c$

Over the available period, we compute the average weights and resulting :

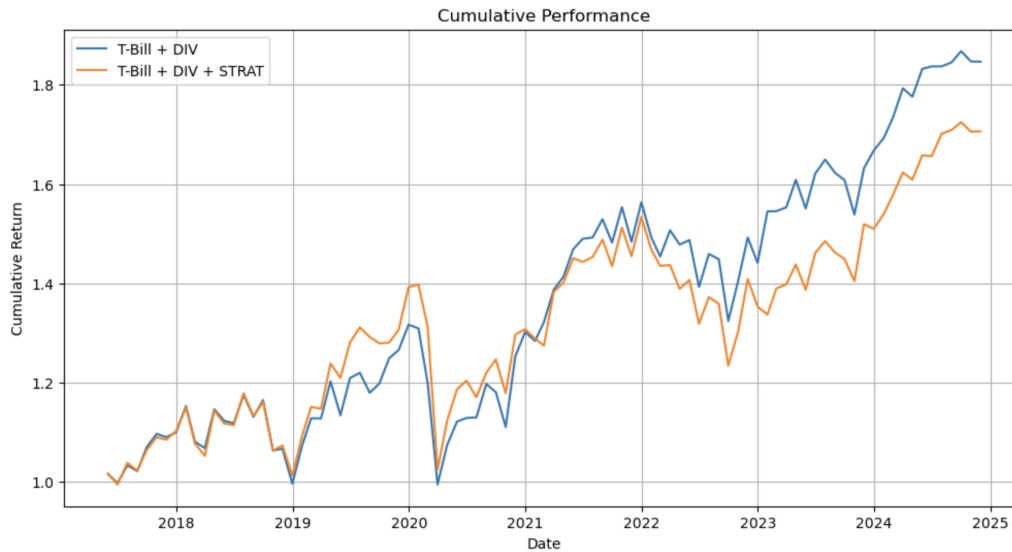
Parameter	b	c	R_{fund}	MOM	REV	CARRY	DOLLAR	T-Bill
Average	0.855	1.500	0.0845	0.207	0.234	0.300	0.758	-1.356

8.d Performance of the strategies

We now compare the two Fund. Surprisingly, the second fund (with overlay) does not outperform the first, despite the expectation that diversification should increase the Sharpe ratio and thus improve returns under a fixed volatility. To investigate this, we computed the correlation matrix of the four overlay components (see Figure 11) and confirmed they are weakly correlated, which in theory should enhance diversification benefits.

A possible explanation lies in the implementation: the estimation of the weights b and c relies on past rolling windows, and we apply them with a one-period shift to avoid look-ahead bias. While unshifted results showed more coherent and stronger performance, we chose the shifted implementation for theoretical consistency. This shift may lead to estimation errors or delayed reaction to market dynamics, weakening the effectiveness of the

strategy. Additional sources of underperformance could include estimation noise in the rolling mean and covariance matrices, or the possibility that the STRAT component has insufficient return despite its low correlation and volatility.



9 Performance and risk analysis for the Fund strategy

9.a

The Fama-French 5 factors model is an asset pricing model that regroups the following 5 factors :

- Market Risk Premium (MKT): The excess return of the market portfolio over the risk-free rate
- Size (SMB - Small Minus Big): The return difference between small-cap and large-cap stocks. Small companies historically outperform large ones.
- Value (HML - High Minus Low): The return difference between high book-to-market (value) stocks and low book-to-market (growth) stocks.
- Profitability (RMW - Robust Minus Weak): The return difference between stocks with high operating profitability and those with low operating profitability.
- Investment (CMA - Conservative Minus Aggressive): The return difference between stocks of companies with low investment (conservative) and high investment (aggressive) behavior.

The times series regression of our new strategy on the Fama-French 5 research factors is given by :

$$R_{FUND}^e = \alpha + \beta_{MKT} R_{MKT}^e + \beta_{SMB} R_{SMB} + \beta_{HML} R_{HML} + \beta_{RMW} R_{RMW} + \beta_{CMA} R_{CMA}$$

We get the following results :

	coef	std err	t	P> t
const	-0.0034	0.003	-1.065	0.290
Mkt-RF	0.7396	0.069	10.649	0.000
SMB	-0.0763	0.132	-0.578	0.565
HML	0.2610	0.116	2.242	0.028
RMW	0.2172	0.160	1.356	0.179
CMA	-0.0045	0.169	-0.027	0.979

Table 1: Regression output

The time-series regression of the strategy's monthly excess returns on the Fama-French five factors delivers

- $\hat{\beta}_{\text{Mkt-RF}} = 0.740$ ($t = 10.6$, $p < 0.001$) and $\hat{\beta}_{\text{HML}} = 0.261$ ($t = 2.24$, $p = 0.028$) — both significant at the conventional 5% level;

- $\hat{\beta}_{\text{SMB}}$, $\hat{\beta}_{\text{RMW}}$, $\hat{\beta}_{\text{CMA}}$ are individually not significant ($|t| < 2$);
- an intercept (abnormal return) $\hat{\alpha} = -0.0034$ with $t = -1.07$ ($p = 0.29$), i.e. statistically indistinguishable from 0;
- $R^2 = 0.651$, so the five factors explain roughly two-thirds of month-to-month return variance.

Hence the strategy behaves like a low-beta, value-tilted equity portfolio; the remaining three factor loadings are small and imprecisely estimated rather than proven to be exactly zero.

9.b

The portfolio has a market beta of ≈ 0.74 , so it moves only three-quarters as much as the broad equity market. It also shows a positive loading on the value factor ($\beta_{\text{HML}} \approx 0.26$). All other Fama–French factor loadings are small and statistically imprecise. The strategy earns two compensated premia:

[label=()]the equity risk premium, scaled down by the 0.74 beta; the value premium, supplied by its tilt toward cheaper stocks.

Those two exposures together explain roughly two-thirds of month-to-month return variation of the strategy ($R^2 = 0.651$).

Efficient-Market Hypothesis (EMH) Under the semi-strong form of the EMH, properly priced risks should leave no systematic abnormal return after factor adjustment. Because $\hat{\alpha}$ is basically zero and the model explains about 65 % of the return variance ($R^2 = 0.651$), the results line up with market efficiency: the strategy’s gains look like fair compensation for taking market- and value-risk, not like profits from mispricing.

Capital Asset Pricing Model (CAPM) The CAPM allows only the market factor. With $\beta_{\text{Mkt-RF}} \approx 0.74$ a single-factor CAPM would predict a lower expected excess return than the full FF-5 model delivers.

The fact that β_{HML} is positive and significant tells us that value stocks add materially to the strategy’s returns. Because the single-factor CAPM has no value term, it would wrongly class this value-driven return as unexplained “alpha.” Hence the CAPM by itself cannot fully describe the strategy.

Arbitrage Pricing Theory (APT) APT posits that, in the absence of arbitrage, expected returns are linear in a **set** of risk factors; the intercept should be zero once all relevant factors are included. Here, adding the full FF-5 drives $\hat{\alpha}$ to insignificance and lifts R^2 to a respectable 0.651. That outcome is fully consistent with the APT: the strategy’s performance is the sum of priced exposures—chiefly market and value—while idiosyncratic bets average out.

10 Appendix

Cumulative Return Comparison: Hedged vs Unhedged Strategies

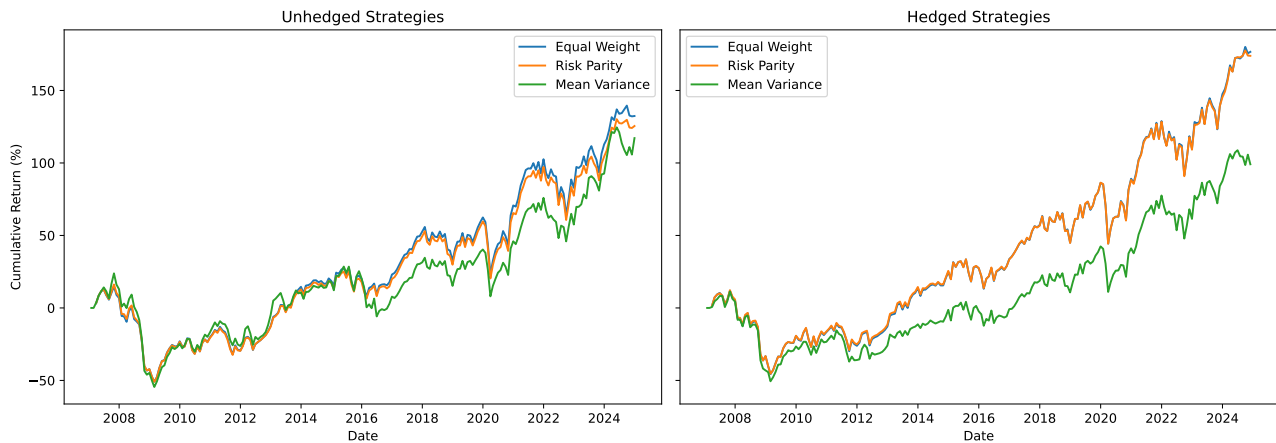


Figure 6: Cumulative performance of Equal Weight, Risk Parity, and Mean-Variance portfolios, with and without currency hedging. The hedged RP strategy (DIV) grows steadily over time. Unhedged strategies are more volatile, particularly EW.

CARRY Returns

Date	Carry Return
2002-02-28	3.77%
2002-03-31	4.88%
2002-04-30	7.37%
2002-05-31	5.33%
2002-06-30	-0.74%

Table 2: First five monthly returns of the CARRY strategy

Interest Rate Differentials and Carry Signal

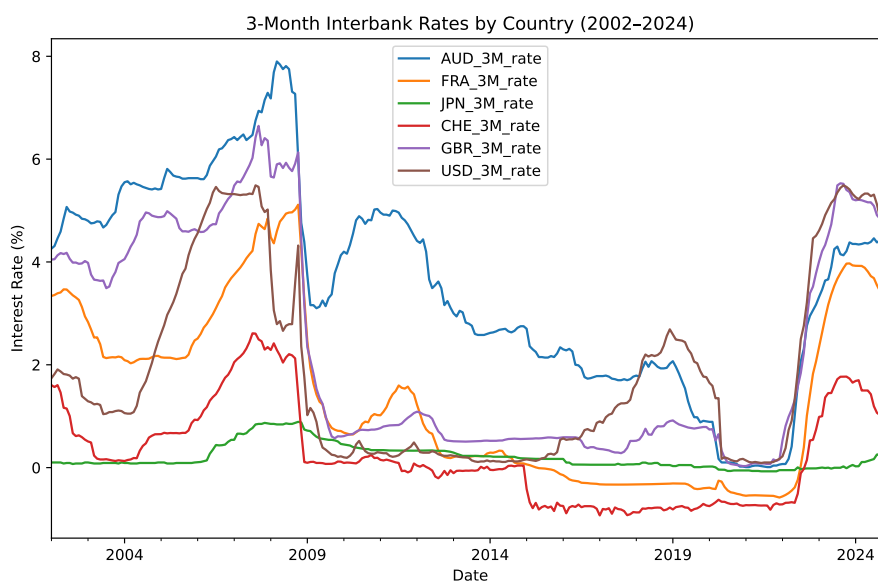


Figure 7: 3-month interbank rates by country from 2002 to 2024. The carry signal was strongest before 2008 when interest rate differentials were wide. Post-crisis, global rates converged and flattened, reducing carry opportunities.

DOLLAR Returns

Date	Dollar Return
2002-02-28	-0.5955%
2002-03-31	0.0447%
2002-04-30	-0.6643%
2002-05-31	-0.2486%
2002-06-30	0.3233%

Table 3: First five monthly returns of the DOLLAR strategy

Illustrations for part 7

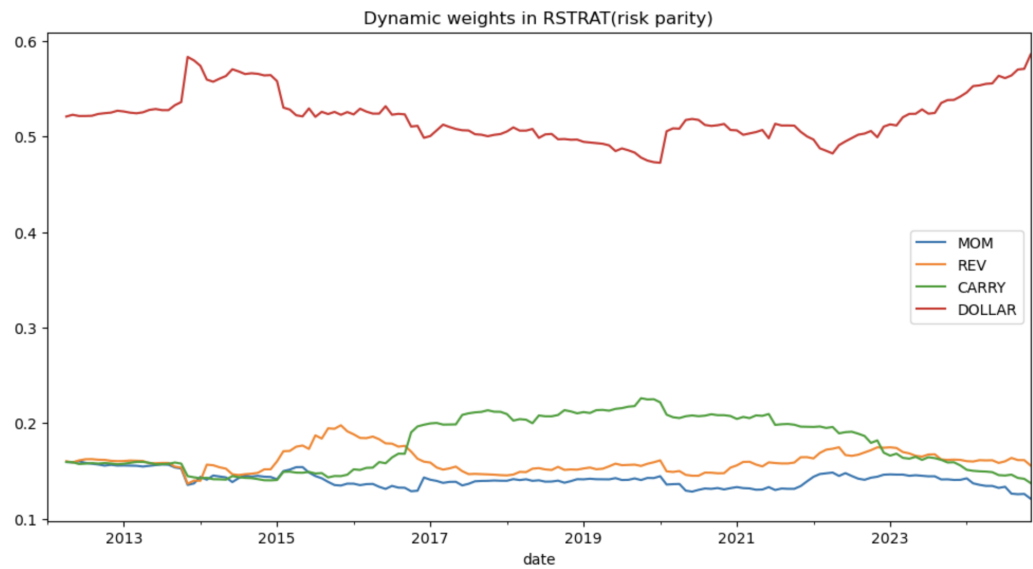


Figure 8

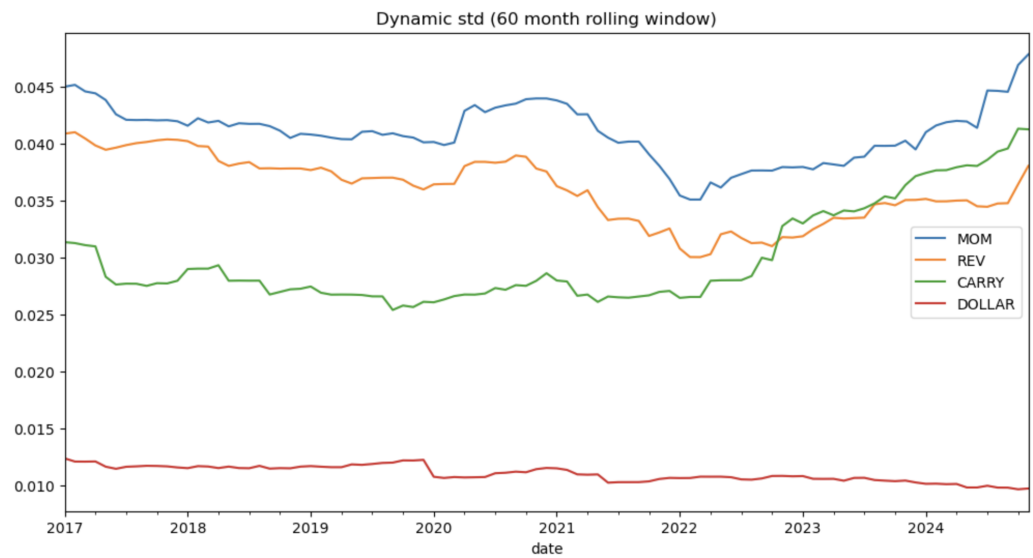


Figure 9

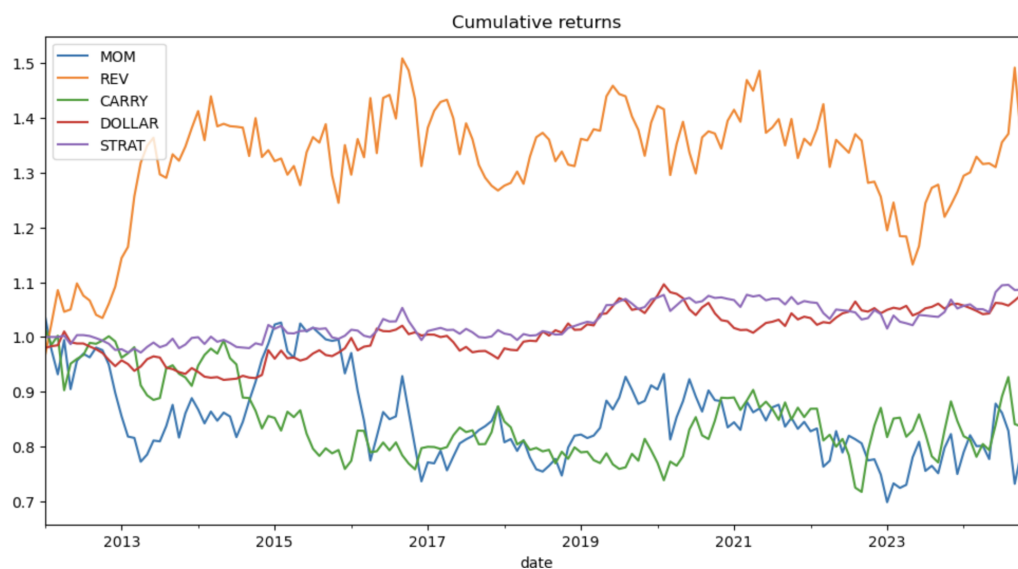


Figure 10

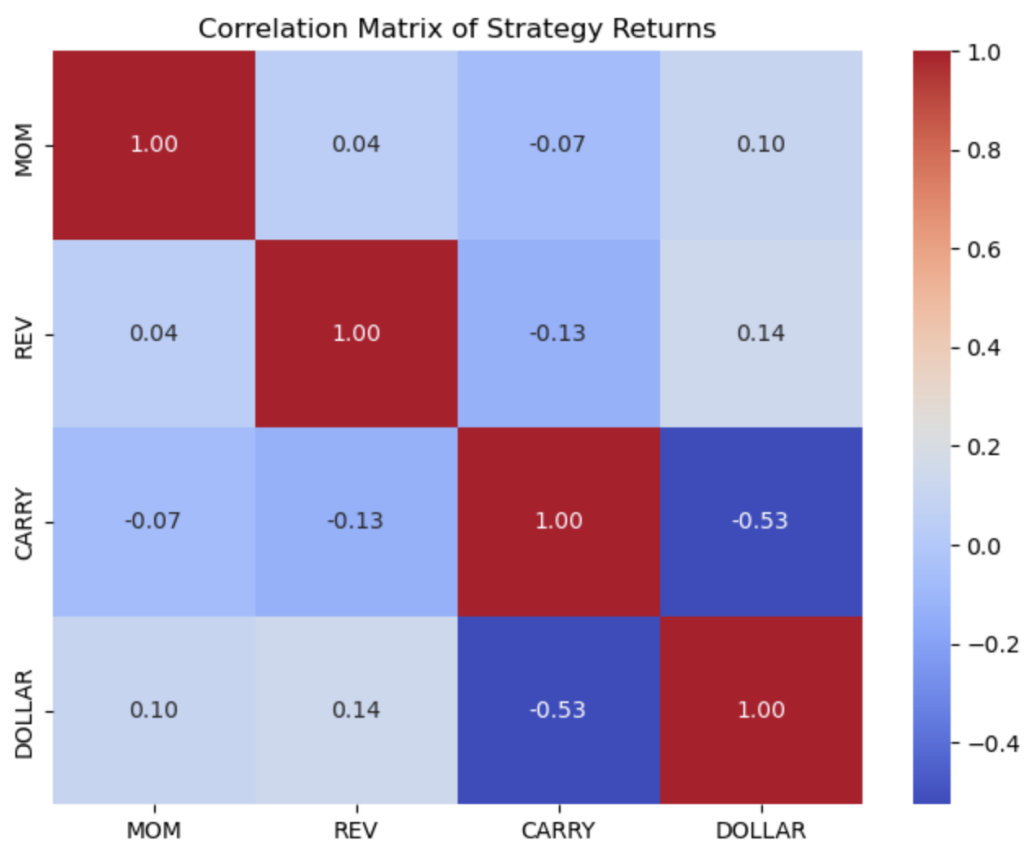


Figure 11